

Smart Voice Control

1. Function Description

Control the robotic arm to perform different actions through voice commands.

2. Startup and Operation

2.1 Startup

Open the terminal and enter the following command to start:

```
python3 ~/dofbot_voice/scripts/intelligent_voice_ctrl.py
```

2.2 Operation Steps

After the program runs, the voice module will broadcast "I'm here", then say "Hi, yahboom" to the voice module. The voice module replies "I'm here" indicating successful wake-up. Then say "Applaud" to the voice module. The voice module will broadcast "OK", and then the gripper of the robotic arm will clamp and release, clamp and release, simulating a clapping action. The voice commands that this program can recognize are as follows:

Voice Command	Effect
ACTION-A	Gripper opens and closes (clapping)
ACTION-B	Robotic arm moves up, servo 4 moves up and down (nodding)
ACTION-C	Robotic arm presents praying posture (praying)
ACTION-D	Robotic arm presents kneeling posture (kneeling)
RESET	Return to robotic arm extended state
ACTION-E	Robotic arm extends upward, turns left and right (startled)

3. Core Code Analysis

Source code path: ~/dofbot_voice/scripts/intelligent_voice_ctrl.py

```
# Import robotic arm low-level control library
import Arm_Lib
# Import voice recognition and broadcast library
from Speech_Lib import Speech

# Create voice recognition and broadcast object mySpeech
mySpeech = Speech()
# Create robotic arm low-level control object Arm
Arm = Arm_Lib.Arm_Device()

# Call speech_read function to get voice recognition result
result = mySpeech.speech_read()
```

```
# Call Arm_serial_servo_write6 function, control six servos to rotate to set  
angles, 1000 means complete the rotation of six servos within 1000 milliseconds  
Arm.Arm_serial_servo_write6(90, 90, 90, 90, 90, 90, 1000)
```